Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The hormone which controls the processes of burning of fats, proteins and carbohydrates to liberate energy in the body is

(A) cortisone

(B) adrenaline

(C) thyroxine

(D) insulin

Q.2

Which one of the following is an amine hormone?

(A) Thyroxine

(B) Oxypurin

(C) Insulin

(D) Progesterone

Q.3

The correct statement in respect of protein haemoglobin is that it

(A) acts as an oxygen carrier in the blood

(B) forms antibodies and offers resistance to diseases

(C) enzymes are specific biological catalysts that can normally function at very high temperature $(\mathrm{T} \sim 1000\ \mathrm{K})$

(D) enzymes are specific biological catalysts that possess well-defined active sites

Q.4

Which is not a true statement?

(A) α -carbon of α -amino acid is asymmetric

(B) All proteins are found in L -form

(C) Human body can synthesize all proteins they need

(D) At pH $= 7$ both amino acids and carboxylic groups exist in the ionised form

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Q.5

Which of the following is a steroid hormone?

(A) Cholesterol

(B) Adrenaline

(C) Thyroxine

(D) Progesterone

Q.6

All of the following statements apply to proteins except

(A) Proteins generally have no definite melting point

(B) Proteins contain the grouping $-\mathrm{CONH}$ -

(C) Proteins have high molecular weight

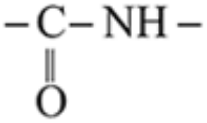
(D) Proteins can only contain the elements C, H, O and N.

Q.7

Which statement is incorrect about peptide bond?

(A) $\mathrm{C}-\mathrm{N}$ bond length in proteins is longer than usual $\mathrm{C}-\mathrm{N}$ bond length

(B) Spectroscopic analysis shows planar structure of



bond

(C) C-N bond length in proteins is smaller than usual $\mathrm{C}-\mathrm{N}$ bond length

(D) None of these

Q.8

A compound of mol wt. 180 is acetylated to give a compound of mol. wt. 390 . The number of amino groups in the initial compound is

(A) 2

(B) 4

(C) 5

(D) 6

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Q.9

Proteins are built up of

(A) Dicarboxylic acids

(B) Amino acids

(C) Alcohols

(D) Hydroxy acids

Q.10

The secondary structure of a protein refers to

(A) $\alpha-$ helical backbone

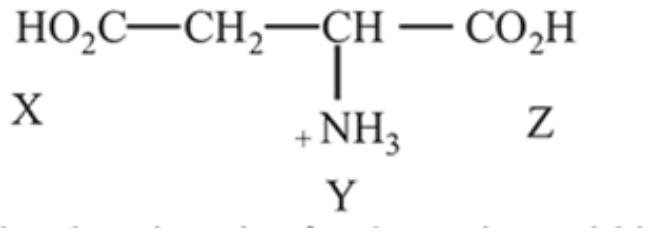
(B) Hydrophobic interactions

(C) Sequence of α -amino acids

(D) Fixed configuration of the polypeptide backbone

Q.11

The pK_a values for the three ionizable groups X, Y and Z of glutamic acid are $4.3, 9.7$ and 2.2 respectively



The isoelectric point for the amino acid is:

(A) 7.00

(B) 3.25

(C) 4.95

(D) 5.95

Q.12

Which of the following statement(s) is (are) true?
(i) All amino acids contain one chiral center.
(ii) Some amino acids contain one, while some contain more chiral center or even no chiral center.
(iii) All amino acids found in proteins have L configuration.
(iv) All amino acids found in proteins have 1° amino group.

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(A) (ii), (iii) & (iv)

(B) (ii) & (iii)

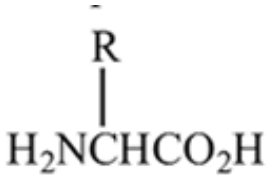
(C) (i), (iii) & (iv)

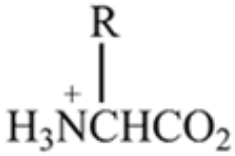
(D) (i) & (iv).

Q.13

Assume that a particular amino acid has an isoelectric point of 6.0. In a solution at $\mathrm{pH} 1.0$ which of the following species will predominate?

(A) 

(B) 

(C) 

(D) 

Q.14

A mixture of two amino acids having pI 9.60 and 5.40 can be separated

(A) by adjusting the pH of the solution at 9.60

(B) by adjusting the pH of the solution at 4.20

(C) by adjusting the pH of the solution at 7.0

(D) by adjusting the pH of the solution at 7.5.

Q.15

Imino acid among these compounds is

(A) serine

(B) proline

(C) tyrosine

(D) lysine

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Q.16

Which amino acid is achiral?

(A) alanine

(B) valine

(C) proline

(D) None of these

Q.17

Iso-electric point of alanine is $(\mathrm{pH} = 6)$. At which pH, maximum concentration of zwitter ion of alanine will be present?

(A) $\mathrm{pH} > 6$

(B) $\mathrm{pH} < 6$

(C) $\mathrm{pH} = 6$

(D) $\mathrm{pH} = 7$

Q.18

$$H{-}C{\equiv}C{-}H \xrightarrow[H_2SO_4]{HgSO_4} (A) \xrightarrow[(2)\ H_3O^+]{(1)\ NH_3 + HCN} (B);$$

Product (B) of given reaction is:

(A) Glycine

(B) Alanine

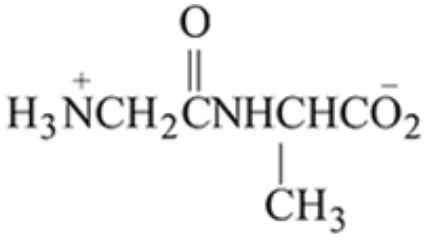
(C) valine

(D) Leucine

Q.19

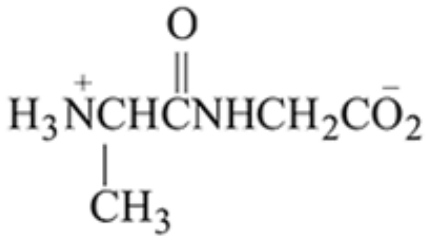
The dipeptide, Gly. Ala has structure -

(A)

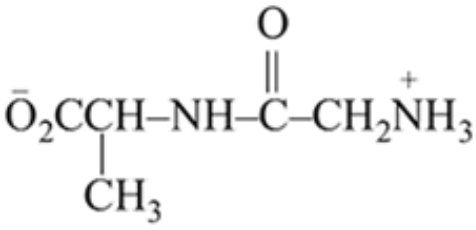


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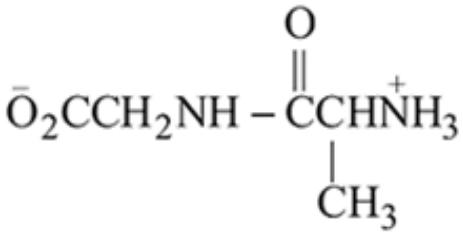
(B)



(C)



(D)



Q.20

What is the monomer of polypeptide

(A) Amino acid

(B) Glucose

(C) Nucleoside

(D) Nucleotide

Answers & Solutions

Q.1 Answer:

thyroxine

Solution:

Thyroxine is the hormone that helps in the regulation of the metabolism of the body i.e., burning of fats, proteins, and carbohydrates.
Adrenaline is released in extreme conditions like - fear, etc.
Insulin helps in the conversion of glucose to glycogen.
Cortisol is one of the steroid hormones which help in a stressful situation.

Q.2 Answer:

Thyroxine

Solution:

Thyroxine is an amine hormone.

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Q.3 Answer:

acts as an oxygen carrier in the blood

Solution:

Haemoglobin acts as an oxygen carrier in the blood since it reacts with oxygen to form unstable oxyhaemoglobin which easily breaks to give back haemoglobin and oxygen.

Q.4 Answer:

All proteins are found in L -form

Solution:

Amino acids are organic compounds that contain amine $(-\mathrm{NH2})$ and carboxyl (-COOH) functional groups, along with a side chain (R group) specific to each amino acid. a -carbon of cc-amino acid is asymmetric. Hence this statement is true. Proteins exist in both D and L-form. So, option (2) is an incorrect statement. The body synthesizes all of its own proteins. The proteins that you obtain through your food are broken down in the digestive system into individual amino acids, which enter the blood stream in the small intestine. From there, they are transported by the blood stream into cells, where they are reassembled into new proteins with the help of ribosomes. This is necessary because proteins are specialized molecules that have to be built in a precise manner in order to perform their function. At $\mathrm{pH}=7$, both amino and carboxylic groups exist in ionised form i.e. zwitter ion.

Q.5 Answer:

Progesterone

Solution:

Progesterone (Gestogens) is a steroid hormone, which controls the development and maintainance of pregnancy. Thryoxine and adrenaline are amine hormones.

Q.6 Answer:

Proteins can only contain the elements C, H, O and N.

Solution:

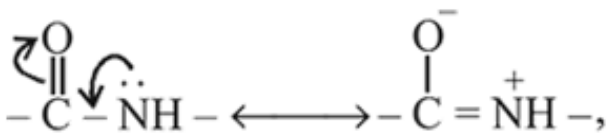
Statement (d) is not correct. Some proteins also contain S, with $\mathrm{C}, \mathrm{H}, \mathrm{O}$ and N.

Q.7 Answer:

$\mathrm{C}-\mathrm{N}$ bond length in proteins is longer than usual $\mathrm{C}-\mathrm{N}$ bond length

Solution:

Due to resonance,



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$C-N$ **bond acquires some double bond character, hence shorter in length**

Q.8 Answer:

5

Solution:

Difference in mass of compound $=390-180=210$
wt. of CH_3CO- **group is** $=43$
Therefore no. of $-NH_2$ **group** $=\frac{210}{43}=4.88=5$

Q.9 Answer:

Amino acids

Solution:

Proteins are polymers of amino acids.
Amino acid $\rightarrow$ **Dipeptide** $\rightarrow$ **Polypeptide** $\rightarrow$ **Protein**

Q.10 Answer:

$\alpha-$ helical backbone

Solution:

The secondary structure of a protein refers to either of α**-helical back bone or** β **-sheet.**

Q.11 Answer:

3.25

Solution:

In acidic medium $\frac{4.3+2.2}{2}=3.25$

Q.12 Answer:

(ii) & (iii)

Solution:

Although D-alanine is a constituent of a bacterial cell walls, it is not found in proteins.

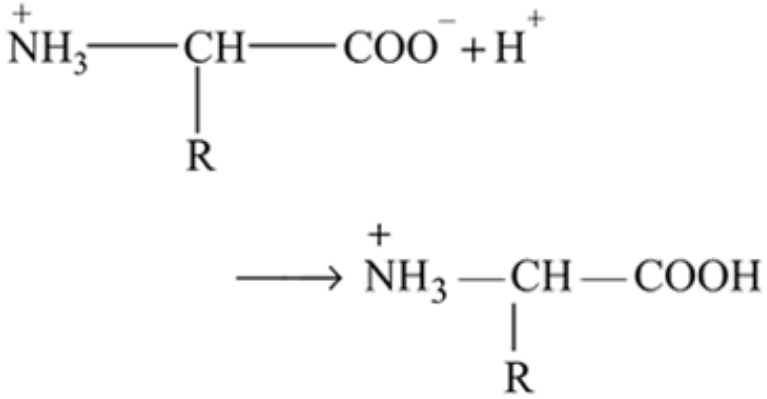
Q.13 Answer:



Solution:

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In acidic medium Zwitter ion convert into



Q.14 Answer:

by adjusting the pH of the solution at 9.60

Solution:

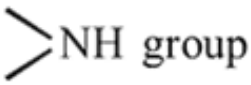
Every amino acid exists exclusively as dipolar ion when the pH of the solution is equal to its isoelectric point (pI), hence at this pH it does not migrate to either electrode, while at other pH, an amino acid migrates either to cathode or to anode depending upon its pI. Thus at $\mathrm{pH}9.60$, amino acid with pI 5.40 will exist as an anion and migrate to anode; while that with pI 9.60 will not migrate to any electrode.

Q.15 Answer:

proline

Solution:

Proline contains imino (secondary amino),



Q.16 Answer:

None of these

Solution:

All are chiral.

Q.17 Answer:

$\mathrm{pH} = 6$

Solution:

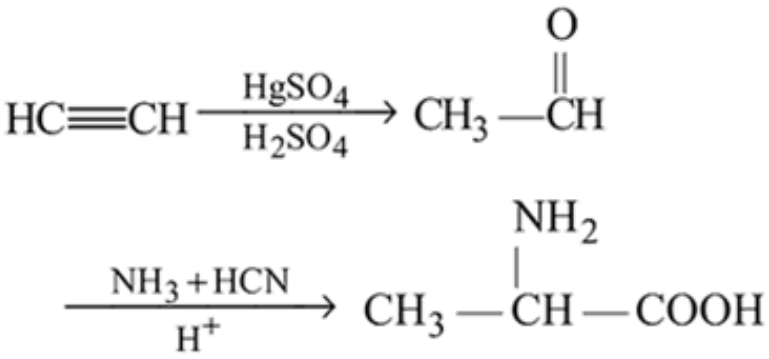
At isoelectric point, conc. of Zwitter ion will be maximum.

Q.18 Answer:

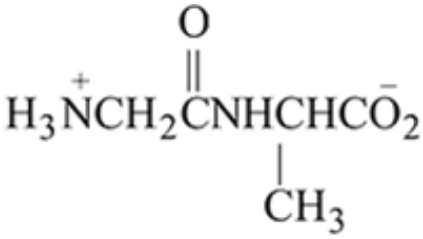
Alanine

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Solution:

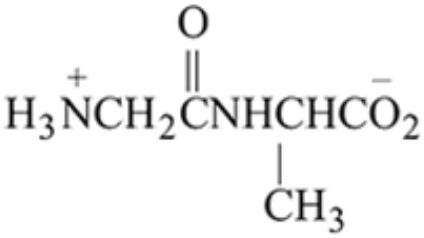


Q.19 Answer:



Solution:

By convention, the amino acid with the free amino group (N-terminal) is written at the left end and the one with the unreacted carboxyl group (C-terminal) at the right end. Thus, the structure of Gly. Ala is



Q.20 Answer:

Amino acid

Solution:

Polypeptide chains are long strings of amino acids held together by peptide bonds.
Amino Acids are linked by Peptide Bonds to form Polypeptide chains. Proteins are linear polymers formed by linking the a-carboxyl group of one amino acid to the α-amino group of another amino acid with a peptide bond (also called an amide bond).

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